

Differentiable unconstrained problems

$$\min: f(x) \quad - \quad \underline{\text{for } f \text{ convex} -}$$

$\hat{x}$ is an optimal solution to this problem if

$$\nabla f(\hat{x}) = 0.$$

f convex

$$f(y) \geq f(\hat{x}) + \langle \nabla f(\hat{x}), y - \hat{x} \rangle \quad \forall y$$

if $\nabla f(\hat{x}) = 0$ above reads:

$$\boxed{f(y) \geq f(\hat{x}) \quad \forall y.}$$

Example: Sum of squares

$$a_1, \dots, a_n$$

$$\text{Find } x \text{ to min: } \frac{1}{n} \sum (a_i - x)^2$$

Take deriv, set $= 0$:

$$-\frac{2}{n} \sum (a_i - x) = 0$$

$$\sum a_i - \sum x = 0$$

$$\sum x = \sum a_i$$

$$n \cdot x = \sum a_i$$

$$\rightarrow \boxed{\frac{1}{n} \sum a_i}$$

Example: Ordinary Least Squares

$$(y_i, x_i), \quad x_i \in \mathbb{R}^d$$

$$\text{OLS: } \frac{1}{n} \sum (y_i - x_i^T \beta)^2$$

$$= \frac{1}{n} \|X\beta - y\|_2^2$$

$$= \frac{1}{n} \left(\beta^T X^T X \beta - 2 \beta^T X^T y + y^T y \right)$$

$$\nabla(\) \doteq \frac{2}{n} \left(X^T X \beta - X^T y + 0 \right) = 0$$

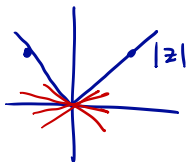
$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T y}$$

Example: Sum of absolute values

$a_1, \dots, a_n$, find x to solve:

$$\min: \frac{1}{n} \sum |a_i - x|$$

$$\hat{x}_{\text{opt.}}: \quad 0 \in \frac{1}{n} \sum \partial(|a_i - \hat{x}|)$$



$$\partial|z| = \begin{cases} -1 & \text{if } z < 0 \\ w, w \in [-1, 1], & \text{if } z = 0 \\ +1 & \text{if } z > 0 \end{cases}$$

Exercise: Apply this definition of subdifferentiation to the above problem to find the solution.

Answer: $x = \text{median}\{a_1, \dots, a_n\}$.

Example: Lasso

Data: $\{(y_i, x_i) : y_i \in \mathbb{R}, x_i \in \mathbb{R}^t\}$.

$$\min: \underbrace{\frac{1}{n} \|X\beta - y\|_2^2}_{\text{RSS}} + \lambda \|\beta\|_1 \quad (\|\beta\|_1 \triangleq \sum |\beta_i|)$$

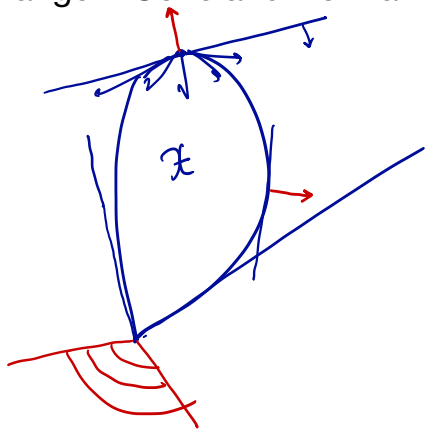
$$\hat{\beta} \text{ optimal if } 0 \in \partial \left(\frac{1}{n} \|X\hat{\beta} - y\|_2^2 + \lambda \|\hat{\beta}\|_1 \right)$$

$$\text{i.e. } 0 \in \frac{2}{n} X^T (X\hat{\beta} - y) + \lambda \partial \|\hat{\beta}\|_1,$$

$$\hat{\beta} \text{ is optimal if } \exists z \text{ with } z_i = \begin{cases} \text{sign}(\hat{\beta}_i) & \text{if } \hat{\beta}_i \neq 0 \\ [-1, 1] & \text{if } \hat{\beta}_i = 0 \end{cases}$$

$$\text{and } \boxed{0 = \frac{2}{n} X^T (X\hat{\beta} - y) + \lambda z}$$

Tangent Cone and Normal Cone



$$T_X(x)$$

$$N_X(x)$$

Optimality conditions for constrained optimization

Tangent cone
at x_2

$$\begin{array}{ll} \min: & f(x) \\ \text{st:} & x \in \mathcal{X} \end{array} \quad (*)$$

f convex
 $\mathcal{X}$ convex

• For x_1 to be optimal need:
 $0 \in \partial f(x_1)$.

• For x_2 to be optimal I need
every direction of allowable movement should have
 $\geq 90^\circ$ angle with $-\nabla f(x_2)$

i.e.,

$$\begin{array}{l} -\nabla f(x_2) \in N_{\mathcal{X}}(x_2) \\ \underline{0 \in \nabla f(x_2) + N_{\mathcal{X}}(x_2)} \end{array}$$

$$\begin{array}{l} \hat{x} \text{ opt. sol'n to } (*) \\ \underline{0 \in \partial f(\hat{x}) + N_{\mathcal{X}}(\hat{x})} \end{array}$$